MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PN, SILICON, UNIJUNCTION
TYPES 2N4947, 2N4948, AND 2N4949
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PN, silicon, unijunction transistors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	P_T 2/ T_C = 25°C	I_e	i_e 3/	V_{B2E}	T_{STG} & T_J
W	W	mA (rms)	A (pk)	V dc	°C
0.36	N/A	50	1	30	-65 to +200

1/ Derate linearly 2.057 mW/°C for T_A > 25°C.
2/ Derate linearly N/A mW/°C for T_C > 25°C.
3/ This value applies for a capacitor discharge through the emitter-base-one diode. The current must fall to 0.37 A within 3 ms and pulse-repetition rate must not exceed 10 pps.

1.4 Primary electrical characteristics at T_A = 25°C.

Type	R_{BBO} V_{B2B1} = 3 V dc I_E = 0		V_{B2B1} = 10 V dc (see figure 5)		$V_{EB1(sat)}$ 1/ V_{B2B1} = 10 V dc I_E = 50 mA dc	V_{OB1} (see figure 9)
	kΩ				V dc	V (pk)
	Min	Max	Min	Max	Max	Min
2N4947	4	9.1	0.51	0.69	3	3
2N4948	4	12.0	0.55	0.82	3	6
2N4949	4	12.0	0.74	0.86	3	3

Type	$(I_{EB20})_1$ V_{EB2} = -30 V dc I_{B1} = 0	$I_{B2(mod)}$ 1/ V_{B2B1} = 10 V dc I_E = 50 mA dc	$R_{\theta JA}$	$R_{\theta JC}$
	nA dc	mA dc	°C/mW	°C/mW
	Max	Min	Max	Max
2N4947	-5	12	0.486	
2N4948	-5	12	0.486	N/A
2N4949	-5	12	0.486	

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

$I_{B2}(mod)$ - - Modulated interbase current. The resultant base-two (B2) current for specified values of emitter current and interbase voltage.

I_e - - - - - Emitter current, rms.

i_e - - - - - Instantaneous emitter current (capacitor discharge).

I_{EB20} - - - Emitter reverse current (dc), base-one (B1) open circuited.

I_p - - - - - Peak point emitter current. This is the minimum value emitter current for which the slope of the static emitter characteristic curve (see figure 3) is zero for a specified value of interbase voltage.

I_v - - - - - Valley point emitter current. This is the maximum value of emitter current for which the slope of the static emitter characteristic curve (see figure 3) is zero for a specified value of interbase supply voltage and base-two (B2) resistance.

- - - - - (eta) Intrinsic standoff ratio. This is defined by the relationship:

$$\eta = \frac{V_p - V_D}{V_{B2B1}}$$

R_{B2} - - - - Resistance in series with the base-two (B2) lead.

R_{BBO} - - - - Interbase resistance. The resistance measured between base-one (B1) and base-two (B2) with the emitter open circuited.

αR_{BBO} - - - - Interbase resistance temperature coefficient. This is determined by the following formula:

$$\alpha R_{BBO} = \left[\frac{(R_{BBO} @ T_1) - (R_{BBO} @ T_2)}{(R_{BBO} @ 25°C)}\right] \frac{100\%}{T_1 - T_2}$$

V_{BB} - - - - Interbase power supply voltage (dc).

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V_{B2B1} - - - - - Interbase voltage, (dc) voltage from base-two (B2) to base-one (B1).

V_{B2E} - - - - - Voltage (dc) from base-two (B2) to emitter (E).

V_D - - - - - - Emitter diode voltage, for intrinsic standoff ratio test. (0.565 ±0.01 V dc at 50 µA dc, T_A = 25°C.)

V_{EB1} - - - - - Voltage (dc) from emitter to base-one (B1).

$V_{EB1(sat)}$ - - - Emitter saturation voltage. The resultant dc voltage measured between the emitter and base-one (B1) for specified values of emitter current and interbase voltage.

V_{OB1} - - - - - Base-one (B1) peak pulse voltage. The base-one (B1) peak pulse voltage is defined as shown in figure 9. This parameter is a relative indicator of the peak emitter current available for use in firing circuits.

V_P - - - - - - Peak point emitter voltage. The voltage from emitter to base-one (B1) when the peak point emitter current flows for a specified value of interbase voltage.

3.2.1 Symbol. The graphic symbol for the unijunction transistor is shown on figure 4.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold, tin, or solder. Lead finish may be specified in the contract or order (see 6.2) without affecting the qualified product status of the device or applicable JAN marking.

3.4 Marking. Device marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

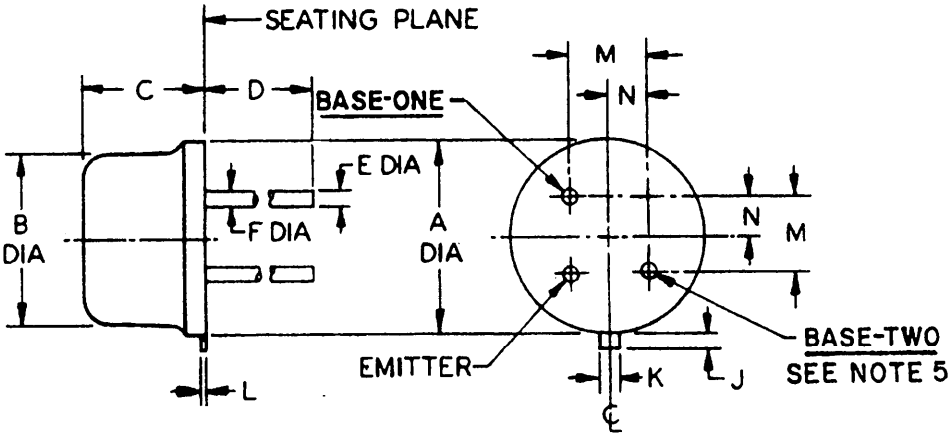
4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
4	See 4.5.9	See 4.5.9
9	$(I_{EB20})_1$, R_{BBO},	Not applicable.
10	See 4.3.1.	See 4.3.1.
11	$(I_{EB20})_1$, R_{BBO}, and ; $\Delta(I_{EB20})_1$ = 100% of initial value or 2 nA dc, whichever is greater; ΔR_{BBO} = ±15% of initial value; Δ = ±10% of initial value.	$(I_{EB20})_1$, R_{BBO}, and
12	See 4.3.2.	See 4.3.2.
13	Subgroups 2 and 3 of group A herein. $\Delta(I_{EB20})_1$ = 100% of initial value or 2 nA dc, whichever is greater; ΔR_{BBO} = ±15% of initial value; Δ = ±10% of initial value.	Subgroup 2 of group A herein. $\Delta(I_{EB20})_1$ = 100% of initial value or 2 nA dc, whichever is greater; ΔR_{BBO} = ±15% of initial value; Δ = ±10% of initial value.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.209	.230	5.31	5.84	
B	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	---	12.70	---	7
E	---	.021	---	.53	2,7
F	.016	.019	.41	.48	3,7
J	.028	.048	.71	1.22	6
K	.036	.046	.91	1.17	
L	---	.020	---	.51	
M	.0707 Nom		1.80 Nom		4
N	.0354 Nom		0.90 Nom		4

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gaging plane .054 $^{+.001}_{-.000}$ (1.37 $^{+.03}_{-.00}$ mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 preferred measuring method.
5. Base-two shall be internally connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All 3 leads.

FIGURE 1. Physical dimensions.

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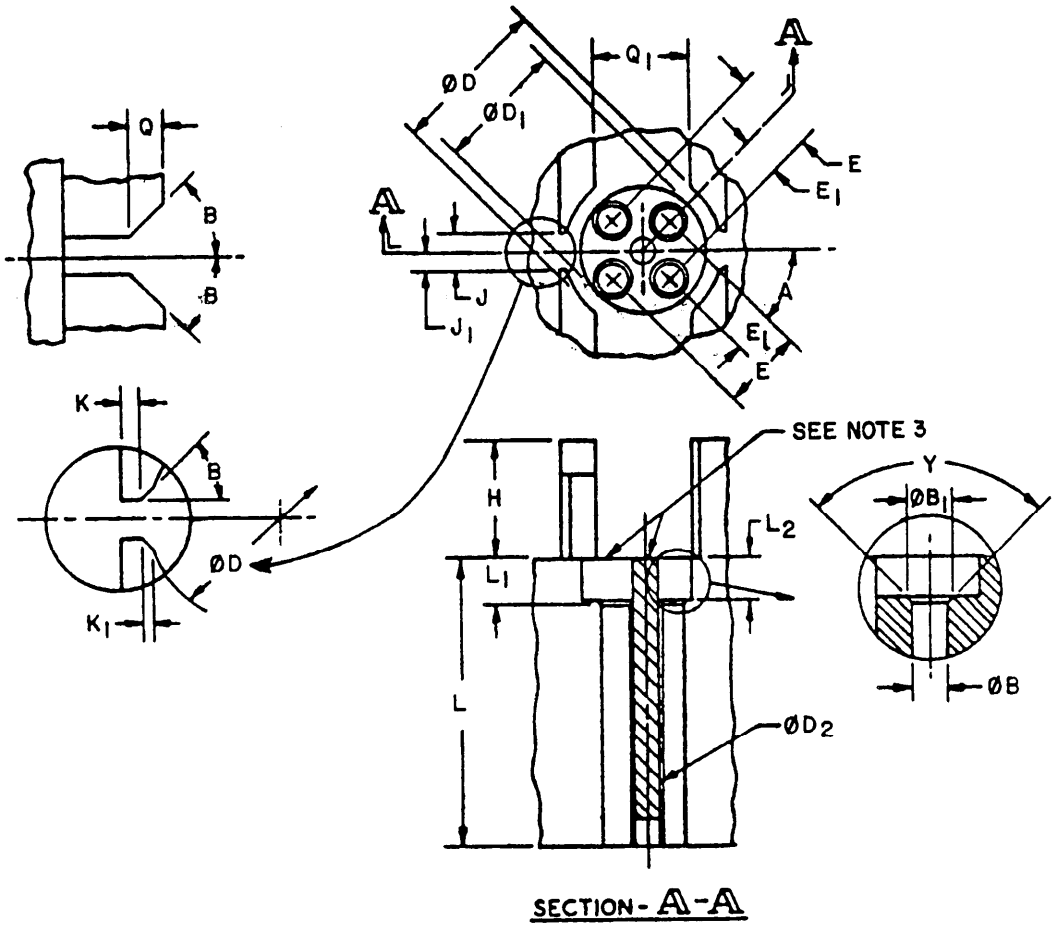


FIGURE 2. Gage for lead and tab location.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ØB	.0325	.0335	.8255	.8509	4
$ØB_1$	.043 Nom		1.09 Nom		4
ØD	.2310	.2315	5.867	5.880	
$ØD_1$	.159	.161	4.04	4.09	
$ØD_2$	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.527	2.553	
E_1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.207	
J_1	.0235	.0245	.597	.622	
K	.009	.011	.229	.279	
K_1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90°	45.10°	---	---	
B	45° Nom		----		
Y	90° Nom		---		

NOTES:
1. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
2. The following gaging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) +.010 (.254 mm) from the seating surface of the gage. A force of 8 ±.5 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage. The use of a pin straightener prior to insertion in the gage is permissible. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (.025 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gage for lead and tab location - Continued.

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4.3.1 High temperature reverse bias test conditions. High temperature reverse bias test conditions shall be as follows: Test method 1039, condition A of MIL-STD-750. Emitter to base-one open circuit.

V_{B2E} = 24 V dc; I_{B1} = 0 for 48 hours minimum at T_A = 150°C (minimum).

4.3.2 Power burn-in test conditions. Power burn-in test conditions shall be as follows:

V_{B2B1} = 10 V dc; I_E = 30 mA dc at T_A = 25°C ±3°C

No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Interbase resistance. The specified interbase voltage (see tables I and IV) is applied and the corresponding resistance is measured for the emitter open circuited.

4.5.3 Intrinsic standoff ratio. Either of the two alternate methods below may be used at the option of the manufacturer.

4.5.3.1 Method a. The circuit of figure 5a shall be used for this measurement. The specified interbase voltage (see tables I and IV) shall be applied and the variable resistor shall be adjusted to obtain a null. Eta is then read directly from the calibration of the variable resistor. The following equation shall be used to calibrate the variable resistor:

$$\eta = \frac{V_P - V_D}{V_{B2B1}}$$

4.5.3.2 Method b. The circuit of figure 5b, or a suitable equivalent, shall be used for this measurement. Switch S1 is closed, applying the specified V_{B2B1} (see tables I and IV) and causing the unijunction to fire. The voltage across the 100 kilohm resistor, V_R, is measured and ratio is computed as follows:

$$\eta = \frac{V_R}{V_{B2B1}} = \frac{V_P - V_D}{V_{B2B1}} \quad (V_D = 0.565 \text{ V dc at } 50\ \mu\text{A dc and } T_A = 25°C)$$

4.5.4 Modulated interbase current and emitter saturation voltage. The specified values (see tables I and IV) of interbase voltage and emitter current are applied. The base-two (B2) current is measured as the modulated interbase current. The emitter to base-one (B1) voltage is measured as the emitter saturation voltage. The test circuit of figure 6 may be used to measure these parameters.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD JANS	LTPD JAN JANTX JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 1			15 (c=0)	5				
Visual and mechanical examination	2071							
Subgroup 2			3 1/	5				
Intrinsic standoff ratio	---	V_{B2B1} = 10 V dc (see 4.5.3)						
2N4947						0.51	0.69	---
2N4948						0.55	0.82	---
2N4949						0.74	0.86	---
Interbase resistance	---	V_{B2B1} = 3 V dc; I_E = 0 (see 4.5.2)			R_{BBO}			
2N4947						4	9.1	kΩ
2N4948						4	12.0	kΩ
2N4949						4	12.0	kΩ
Modulated interbase current	---	V_{B2B1} = 10 V dc (see 4.5.4); I_E = 50 mA dc Pulsed (see 4.5.1)			$I_{B2(mod)}$	12	---	mA dc
Emitter saturation voltage	---	V_{B2B1} = 10 V dc (see 4.5.4); I_E = 50 mA dc Pulsed (see 4.5.1)			$V_{EB1(sat)}$	---	3	V dc
Emitter reverse current	3041	Bias cond D (see 4.5.7); V_{EB2} = -30 V dc; I_{B1} = 0			$(I_{EB20})_1$	---	-5	nA dc
Valley point emitter current	---	V_{BB} = 20 V dc; R_{B2} = 100 ohms (see 4.5.6)			I_{V1}			
2N4947						4	---	mA dc
2N4948						2	---	mA dc
2N4949						2	---	mA dc
Peak point emitter current	---	V_{B2B1} = 25 V dc (see 4.5.5)			I_P			
2N4947						---	2	µA dc
2N4948						---	2	µA dc
2N4949						---	1	µA dc
Base-one peak pulse voltage	---	See figure 9			V_{OB1}			
2N4947						3	---	V pk
2N4948						6	---	V pk
2N4949						3	---	V pk
Subgroup 3				5				
Interbase resistance temperature coefficient	---	V_{B2B1} = 3 V dc; I_E = 0 (see 4.5.8); T_1 = 100°C; T_2 = -65°C			αR_{BBO}	0.1	0.9	%/°C
High temperature operation:		T_A = 125°C						
Emitter reverse current	3041	Bias cond D (see 4.5.7) V_{EB2} = -30 V dc; I_{B1} = 0			$(I_{EB20})_2$	---	-1	µA dc
Valley pointer emitter current	---	V_{BB} = 20 V dc; R_{B2} = 100 ohms (see 4.5.6)			I_{V2}			
2N4947						3.0	---	mA dc
2N4948						1.5	---	mA dc
2N4949						1.5	---	mA dc
Subgroups 4, 5, 6, 7								
Not applicable								

1/ For JANS level, all devices required by the specified LTPD shall be subjected to subgroups 2 and 3 combined.

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TABLE IIa. Group B inspection for JANS devices.

Examination or test	MIL-STD-750 Method	MIL-STD-750 Conditions	Qualification and large lot quality conformance inspection LTPD	Small lot quality conformance inspection n/c
Subgroup 1			10	8/0
Physical dimensions	2066	(See figure 1)		
Subgroup 2			15	6/0
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3			10	6/0
Thermal shock (temperature cycling)	1051			
Hermetic seal a. Fine b. Gross	1071			
Electrical measurements		See table IV, steps 1, 2, 3, 4, and 5		
Decap internal visual design verification	2075		6/0	
Bond strength	2037	Test cond A All internal wires for each device shall be pulled separately.		
Die shear	2017	Devices previously subjected to the bond strength test may be used for this test.		
Subgroup 4			10	N/A
Intermittent operation life	1037	V_{B2B1} = 10 V dc; I_F = 30 mA dc at T_A = 25°C ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2000 cycles. No heat sink or forced air on the devices shall be permitted.		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, and 10		
Subgroup 5			10	12/2
Steady-state operation life (accelerated)	1027	V_{B2B1} = 10 V dc; I_F = 30 mA dc at T_A = 100°C for 96 hours. Marking legibility requirements shall not apply.		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 7, 8, and 9		
Subgroup 6				
Not applicable				
Subgroup 7			5	12/0
Constant acceleration	2006	One minute minimum in the Y_1 orientation at 30,000G minimum. Only required when lot complies with 4.5.9.		
Electrical measurements		See table IV, steps 1, 2, 3, 4, and 5		

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Examination or test	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD
Subgroup 1			15
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			10
Thermal shock (temperature cycling)	1051		
Hermetic seal a. Fine leak b. Gross leak	1071		
Electrical measurements		See table IV, steps 1, 2, and 5	
Subgroup 3			5
Steady-state operation life	1027	V_{B2B1} = 10 V dc, I_F = 30 mA dc at T_A = 25°C ±3°C. No heat sink or forced air cooling on the devices shall be permitted.	
Electrical measurements		See table IV, steps 1, 2, and 6	
Subgroup 4 1/			
Decap internal visual design verification	2075		1 device/0 failure for each lot
Bond strength	2037	Test cond A	20 (c = 0)
Subgroup 5			
Not applicable			
Subgroup 6			7
High-temperature life (non-operating)	1032	T_A = 200°C	
Electrical measurements		See table IV, steps 1, 2, and 6	
Subgroup 7			5
Constant acceleration	2006	One minute minimum in the Y_1 orientation at 30,000G minimum. Only required when lot complies with 4.5.9	
Electrical measurements		See table IV, steps 1, 2, and 5	

1/ Electrical reject devices from the same inspection lot may be used for all subgroups when electrical end point measurements are not required.

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TABLE III. Group C inspection (all quality levels).

Examination or test	MIL-STD-750 Method	Conditions	LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 1			15				
Physical dimensions	2066	(See figure 1)					
Subgroup 2			10				
Thermal shock (glass strain)	1056	Test cond A					
Terminal strength	2036	Test cond E					
Hermetic seal a. Fine leak b. Gross leak	1071						
Moisture resistance	1021						
External visual	2071						
Electrical measurements		See table IV, steps 1, 2, and 5 (JAN, JANTX, JANTXV) and steps 1, 2, 3, 4, and 5 (JANS)					
Subgroup 3			10				
Shock	2016						
Vibration, variable frequency	2056						
Constant acceleration	2006						
Electrical measurements		See table IV, steps 1, 2, and 5 (JAN, JANTX, JANTXV) and steps 1, 2, 3, 4, and 5 (JANS)					
Subgroup 4			15				
Salt atmosphere (corrosion)	1041						
Subgroup 5							
Not applicable							
Subgroup 6			λ=10				
Steady-state operation life	1026	V_{B2B1} = 10 V dc, I_E = 30 mA dc at T_A = 25°C ±3°C. No heat sink or forced air cooling on the devices shall be permitted.					
Electrical measurements		See table IV, steps 1, 2, and 6 (JAN, JANTX, JANTXV) and steps 1, 2, 3, 4, 5, 7, 8, and 9 (JANS)					

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TABLE IV. Groups B and C electrical measurements.

Step	Examination or test	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1	Intrinsic standoff ratio	---	V_{B2B1} = 10 V dc (see 4.5.3)				
	2N4947				0.51	0.69	---
	2N4948				0.55	0.82	---
	2N4949				0.74	0.86	---
2	Interbase resistance	---	V_{B2B1} = 3 V dc; I_E = 0 (see 4.5.2)	R_{BB0}			
	2N4947				4	9.1	kΩ
	2N4948				4	12.0	kΩ
	2N4949				4	12.0	kΩ
3	Modulated interbase current	---	V_{B2B1} = 10 V dc (see 4.5.4) I_E = 50 mA dc Pulsed (see 4.5.1)	$I_{B2(mod)}$	12	---	mA dc
4	Emitter saturation voltage	---	V_{B2B1} = 10 V dc (see 4.5.4) I_E = 50 mA dc Pulsed (see 4.5.1)	$V_{EB1(sat)}$	---	3	V dc
5	Emitter reverse current	3041	Bias cond D (see 4.5.7) V_{EB2} = -30 V dc; I_{B1} = 0	$(I_{EB20})_1$	---	5	nA dc
6	Emitter reverse current	3041	Bias cond D (see 4.5.7) V_{EB2} = -30 V dc; I_{B1} = 0	$(I_{EB20})_1$	---	10	nA dc
7	Intrinsic standoff ratio	---	V_{B2B1} = 10 V dc (see 4.5.3)	Δ		±15% of initial value	
8	Interbase resistance	---	V_{B2B1} = 3 V dc; I_E = 0 (see 4.5.2)	ΔR_{BB0}		±20% of initial value	
9	Emitter reverse current	3041	Bias cond D (see 4.5.7) V_{EB2} = -30 V dc; I_{B1} = 0	$\Delta(I_{EB20})_1$	---	100% of initial value or 4 nA dc, whichever is greater	
10	Emitter saturation voltage	---	V_{B2B1} = 10 V dc (see 4.5.4) I_E = 50 mA dc Pulsed (see 4.5.1)	$\Delta V_{EB1(sat)}$	---	±50 mV dc change from initial value	

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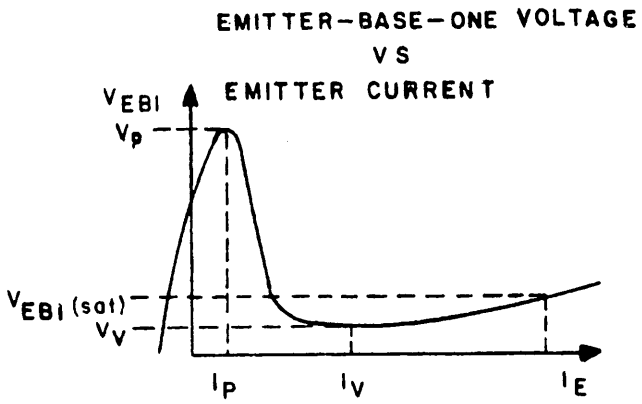


FIGURE 3. Unijunction transistor static emitter characteristic curve.

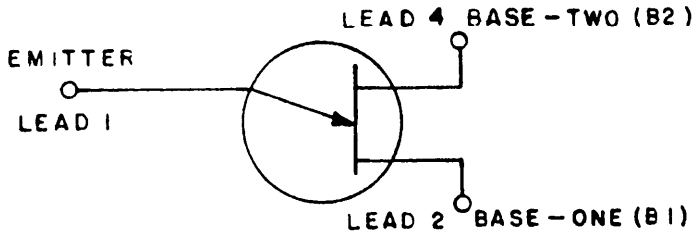
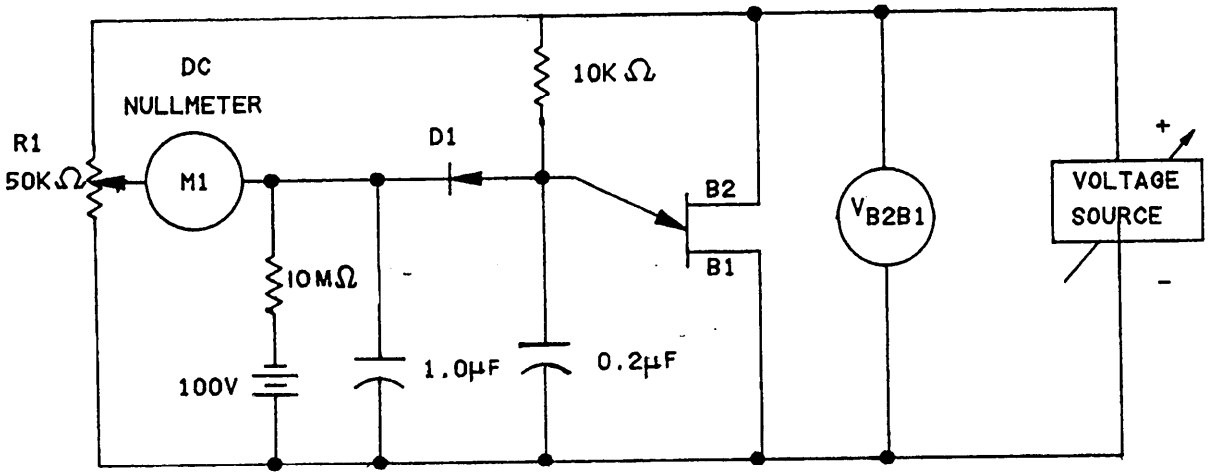


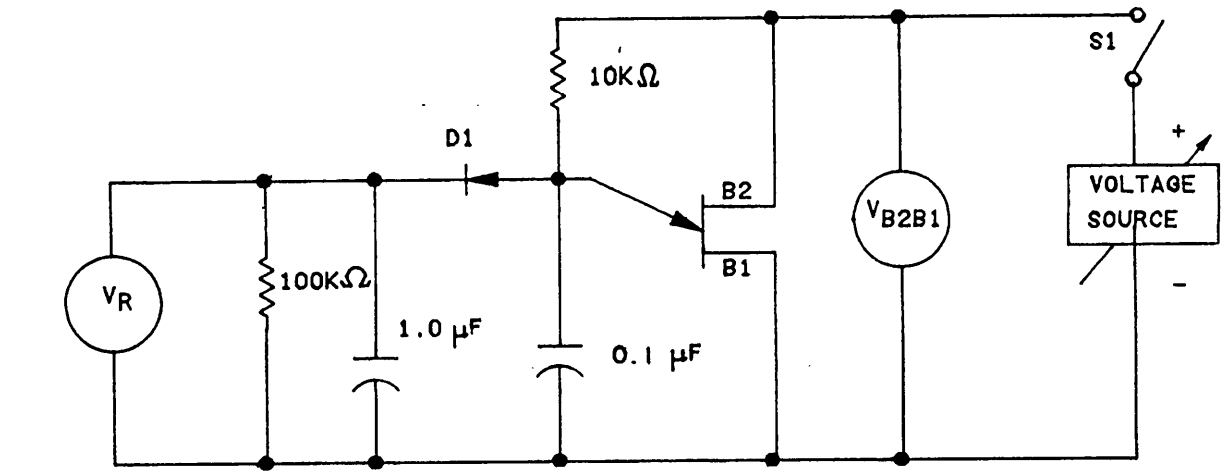
FIGURE 4. Graphic symbol for the unijunction transistor (see 3.2.1).

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NOTES:
1. R1 - Calibrated helipot (n).
2. Voltage source shall have less than 10 mV ripple (peak to peak).
3. D1 - Silicon diode, forward voltage of .565 ±.010 volt at forward current of 50 μA dc and a reverse leakage of less than 1 μA dc at 20 volts.

FIGURE 5a. Intrinsic standoff ratio test circuit (method a).



NOTES:
1. Voltage source shall have less than 10 mV ripple (peak to peak) and be readily controllable to less than 10 mV.
2. D1 - Silicon diode, forward voltage of .565 ±.010 volt at forward current of 50 μA dc and a reverse leakage of less than 1 μA dc at 20 volts.

FIGURE 5b. Intrinsic standoff ratio test circuit (method b).

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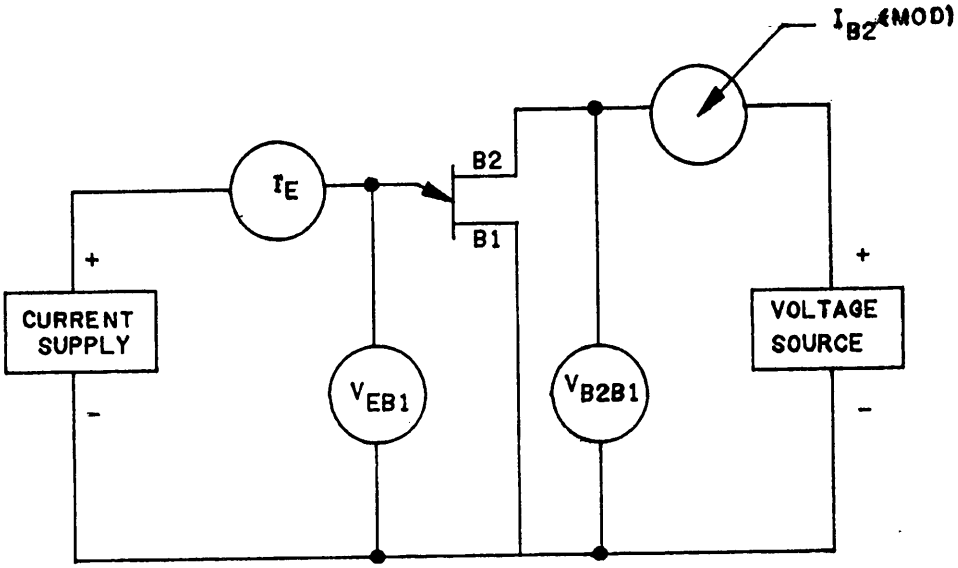
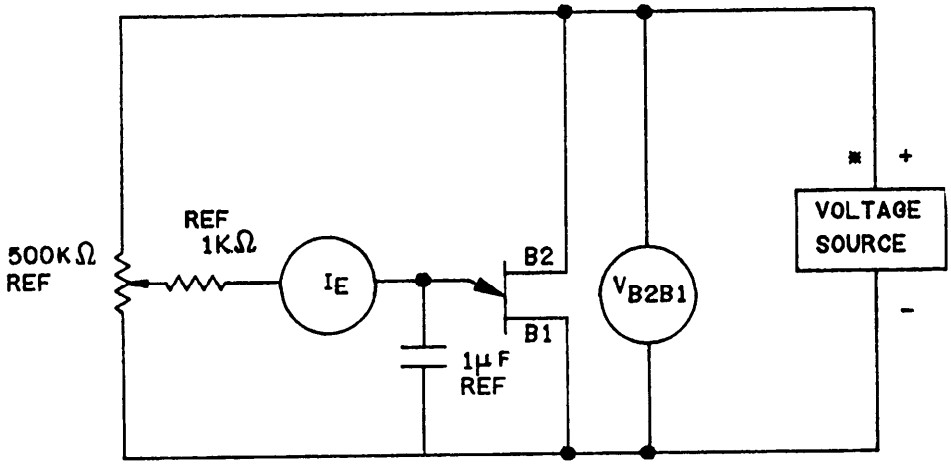


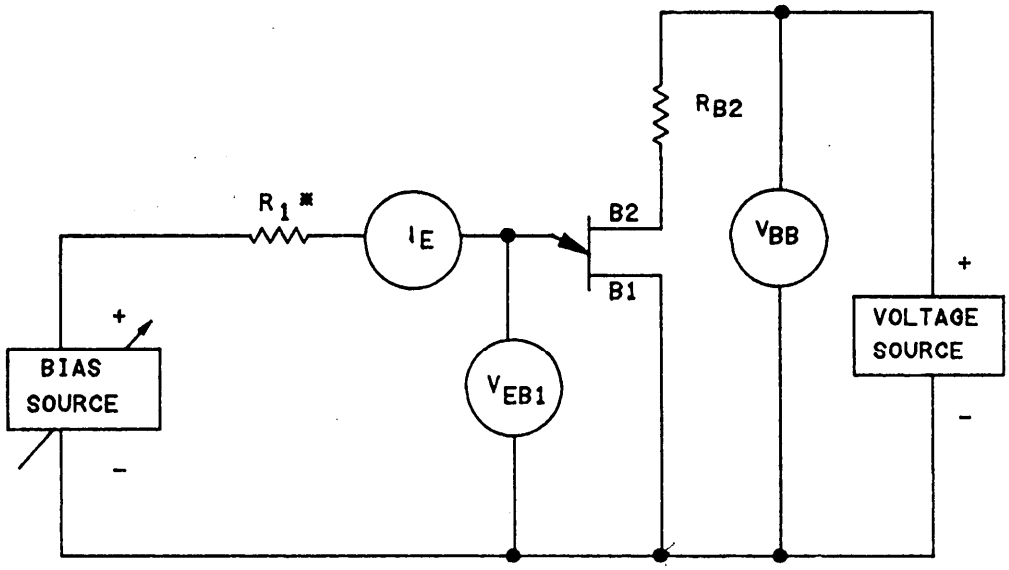
FIGURE 6. Modulated interbase current and emitter saturation voltage circuit.



NOTES:
1. 500 kΩ potentiometer must be noiseless to prevent false triggering.
2. *Voltage source with less than 1 mV peak-to-peak ripple.

FIGURE 7. Peak point emitter current circuit.

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*R_1 chosen to limit current to a safe value.

FIGURE 8. Valley point emitter current circuit.

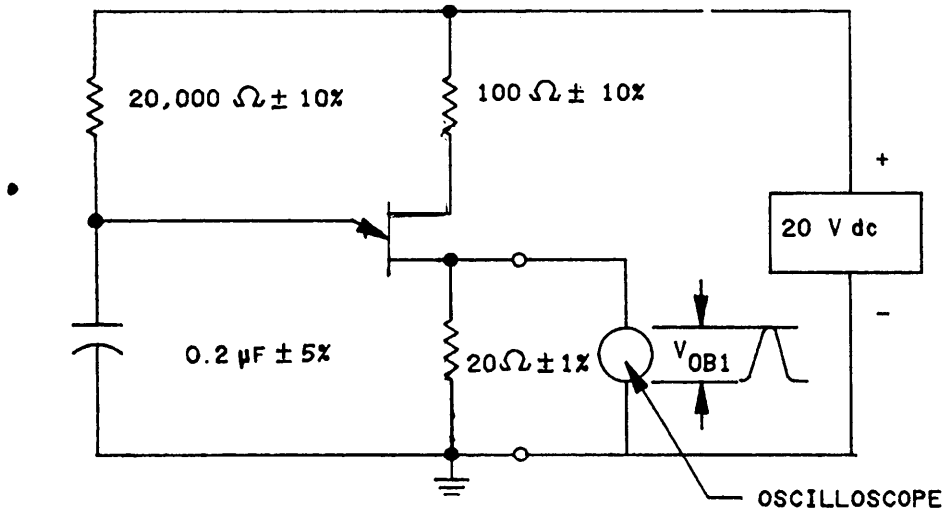


FIGURE 9. Base-one (B1) peak pulse voltage test circuit.

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4.5.5 Peak point emitter current. This parameter shall be measured in the circuit of figure 7 or a suitable equivalent. The potentiometer is adjusted to apply a low resistance between the emitter and base-one (B1). The specified V_{B2B1} (see table I) is applied and the resistance between the emitter and base-one (B1) is increased until the transistor fires as determined by oscillation of I_E. Peak point emitter current is the maximum value of I_E just prior to oscillation.

4.5.6 Valley point emitter current. For the specified interbase supply voltage (see table I) and base-two (B2) resistance, the emitter current corresponding to the valley point operating condition is measured. The test circuit of figure 8, or suitable equivalent, shall be used for this measurement. The specified interbase supply voltage, V_{BB}, and base-two (B2) series resistance shall be applied. R_1 shall be of sufficient value to adequately control the emitter current. The bias voltage shall be gradually increased until the device fires and then shall be varied to obtain a minimum value of V_{EB1}. The I_E corresponding to this minimum value of V_{EB1} is the I_V of the device under test.

4.5.7 Emitter reverse current. This test shall be conducted in accordance with method 3041 of MIL-STD-750 except that the words "collector", "base", and "emitter" shall be replaced with "base-two (B2)", "emitter (E)", and "base-one (B1)", respectively.

4.5.8 Interbase resistance temperature coefficient. The interbase resistance temperature coefficient shall be determined by measuring the interbase resistance in accordance with 4.5.2 at the specified temperatures and computing the temperature coefficient by the following formula:

4.5.9 Constant acceleration. Only devices constructed with aluminum bonding wire which successfully pass three consecutive lots with zero failures shall not require this test as a 100 percent screen on future inspection lots. In the event the sample fails the LTPD requirements of group B inspection, and the failures are catastrophic (i.e., open or shorted condition), as verified by failure analysis, then 100 percent screening shall be re-instituted. The 100 percent screen shall be required on the failed lot and following lots until three successive inspection lots pass the constant acceleration test with zero failures. Only then may sampling inspection be resumed. End point measurements for this screening test shall be those required by screen 11. Only device failures, verified by failure analysis, caused by the constant acceleration test shall be counted as failures for this screening test. All sample devices which are drawn for constant acceleration inspection shall have successfully completed thermal shock inspection (minimum of 20 cycles).

$$\alpha_{R_{BBO}} = \frac{(R_{BBO} \text{ @ } T_1) - (R_{BBO} \text{ @ } T_2)}{(R_{BBO} \text{ @ } 25^\circ C)} \frac{100\%}{T_1 - T_2}$$

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. The contract or purchase order should specify the following:

a. Lead finish if specific finish is required (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - MSFC-EG-02

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 19, 85, 80
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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